Toward a Zero Trust-Based Multi-Layer Security Architecture for Open RAN*

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Abstract

Open Radio Access Network (Open RAN) introduces open interfaces, cloud-native deployment, artificial intelligence (AI)-driven control, and multi-vendor ecosystems. While these characteristics improve flexibility and innovation, they also expand the attack surface and invalidate the implicit trust assumptions of traditional radio access networks. Recent attacks against telecommunications infrastructure demonstrate that conventional perimeter-based security is insufficient for Open RAN environments. This paper proposes a Zero Trust-based multi-layer security architecture that integrates decentralized identity management, broadcast integrity protection, and runtime security monitoring under a unified policy decision framework. The proposed architecture follows the principles of NIST SP 800-207 and enables continuous verification across Open RAN components. The study argues that effective Open RAN security requires coordinated cross-layer enforcement rather than isolated security controls.

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